Speech-Based Pattern Recognition for Emotion Classification Using CREMA-D

Ransford Nyarko
85542025

Dharshani Anandkumar
82653870

***Abstract*—Speech emotion recognition (SER) predicts an emotional label from speech using acoustic cues such as prosody, energy, and spectral structure. In this project, we built an end-to-end SER pipeline on the CREMA-D dataset and emphasized speaker-independent evaluation by splitting the data by speaker to reduce leakage. We implemented and compared (i) a classical baseline using Mel/MFCC summary statistics with Logistic Regression, (ii) a deep spectrogram model using a CNN–BiLSTM on fixed-size log-mel inputs, and (iii) a transfer-learning baseline using pretrained wav2vec 2.0 embeddings with a lightweight classifier. Across experiments, wav2vec 2.0 embeddings generalized better than the CNN–BiLSTM on held-out test speakers, and a simple probability-averaging ensemble of wav2vec 2.0 and CNN–BiLSTM achieved the best overall performance (69.9% validation accuracy and 61.9% test accuracy on our speaker-disjoint split). We report preprocessing choices, model designs, and evaluation results, and we discuss concrete next steps to improve generalization and reduce common confusions observed in the confusion matrix.**

***Keywords*—speech emotion recognition, CREMA-D, speaker-independent split, log-mel spectrogram, CNN–BiLSTM, wav2vec 2.0, ensemble**

I. Introduction

Humans encode emotion in speech through prosody, intensity, speaking rate, pitch dynamics, articulation, and spectral characteristics. Reliable SER enables applications such as assistive technologies, call-center analytics, healthcare screening, affect-aware tutoring, and robust human–computer interaction. SER remains difficult because emotional expression is subtle and confounded by speaker identity, recording conditions, lexical content, and acting style.

A common pitfall in SER evaluation is *speaker leakage*: when the same speaker appears in both training and test splits, models can exploit speaker-specific cues and inflate accuracy. This project therefore prioritizes a *speaker-disjoint* (speaker-independent) split strategy on CREMA-D, a widely used dataset of acted emotional speech containing six target emotions: anger, disgust, fear, happy, neutral, and sad [1].

Contributions. This report summarizes: (i) a reproducible end-to-end preprocessing pipeline for CREMA-D audio-only SER; (ii) a progression of models from classical baselines to deep time–frequency models and transformer-style architectures; (iii) transfer-learning via wav2vec 2.0 embeddings [2]; (iv) an ensemble combining complementary feature spaces; and (v) an error analysis grounded in our confusion matrix with targeted next steps.

II. Related Work

A. Datasets and Evaluation Protocols

CREMA-D provides 7,442 clips from 91 actors and includes crowd-sourced categorical labels and intensity ratings [1]. The dataset paper highlights that audio-only emotion recognition is challenging (human performance around 40.9% on audio-only clips) [1]. Other SER datasets include IEMOCAP [3]. Because SER models can inadvertently learn speaker identity, many SER studies emphasize subject-independent evaluation (speaker-disjoint splits or leave-one-speaker-out) to better approximate deployment on unseen speakers. A concise overview of SER benchmarks and evaluation considerations is provided by Schuller [4].

B. Feature-Based SER

Traditional SER pipelines rely on engineered acoustic features such as MFCCs, pitch, and energy, often summarized using statistics over time. Shallow classifiers (e.g., SVMs, Logistic Regression) can perform competitively but often plateau when temporal dynamics and robust invariances are required.

C. Deep Time–Frequency Models

A common deep-learning approach is to treat a log-mel spectrogram as a time–frequency "image" and learn features with CNNs, then capture longer dynamics with RNNs (CRNN/CNN–LSTM/BiLSTM). For example, CNN–LSTM architectures have been used effectively for SER by combining local spectral pattern learning with temporal aggregation [5]. End-to-end deep models that learn representations directly from low-level inputs have also shown that emotion cues can be captured without heavy feature engineering [6]. Attention mechanisms and learned pooling have been explored to focus on emotionally salient regions within an utterance, improving robustness when emotional cues are sparse [7].

D. Transformers and Self-Supervised Speech Representations

Transformers have been adopted for audio classification by applying self-attention to patchified spectrograms, such as the Audio Spectrogram Transformer (AST) [8]. Self-supervised learning (SSL) has reshaped speech modeling by learning general-purpose embeddings from large amounts of unlabeled audio. wav2vec 2.0 [2] and HuBERT [9] learn representations that transfer well to downstream tasks; benchmarks such as SUPERB evaluate transfer across many speech tasks [10]. For

SER, SSL embeddings often capture prosodic and paralinguistic cues that are difficult to hand-engineer, and can outperform classical features with limited labels, especially when fine-tuned.

III. Dataset and Preprocessing

A. Dataset

CREMA-D is an acted speech emotion recognition dataset comprising short spoken utterances produced under six emotional states: anger, disgust, fear, happiness, neutral, and sadness [1]. Each recording consists of a single sentence and typically lasts approximately 3 seconds. The dataset features performances from 91 actors (48 male, 43 female) aged 20 to 74, representing diverse racial and ethnic backgrounds, including African American, Asian, Caucasian, Hispanic, and others. Utterances are selected from a fixed set of 12 sentences and vary in emotional expression across speakers. For the purposes of this study, only the audio modality is utilized, and the emotion category serves as the classification label [1]. The distribution of samples across classes is presented in Figure 1.

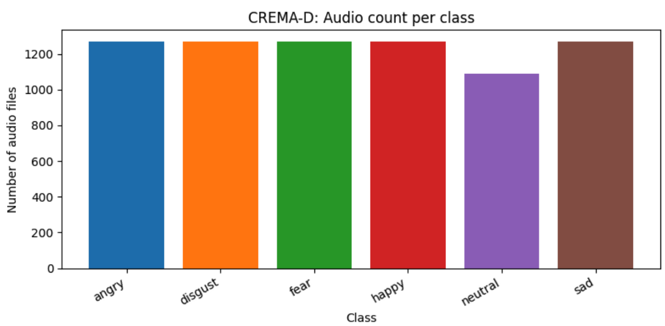

Fig. 1. Dataset distribution across each class.

B. Speaker-Disjoint Splits

To minimize speaker leakage, a speaker-disjoint split is performed using the actor (speaker) ID, ensuring that each speaker appears in only one of the training, validation, or test sets. With 91 speakers in CREMA-D, a 15% speaker-level test split and a 15% validation split of the remaining speakers, results in a 65/12/14 train/val/test speakers (approximately 71%/13%/15% by speaker). This setting is stricter (and typically yields lower scores) than random clip-level splits, but better reflects deployment where the system encounters unseen speakers. We further check class distributions after splitting and find that each emotion is represented in all splits with comparable proportions.

C. Audio Normalization and Features

All clips are resampled to 16 kHz, trimmed to remove leading/trailing silence, and peak-normalized. Resampling standardizes the time scale across recordings and reduces compute, while trimming avoids allocating model capacity to non-speech regions.

We explored two complementary feature families:

TABLE I
KEY PREPROCESSING AND TRAINING SETTINGS.

Component	**Setting**
Sample rate	16 kHz
Log-mel bins	64
Frame sizing	pad/crop to `MAX_FRAMES`
Train crop	random crop when too long
Val/test crop	center crop when too long
Optimizer (CNN–BiLSTM)	Adam, lr 10^{-3}
Weight decay	10^{-4}
Batch size	32
Early stopping	patience 30 (max 80 epochs)
Ensemble weight	$\alpha = 0.25$

(1) Log-mel spectrograms. We compute log-mel spectrograms with 64 mel bins and $f_{\min} = 50$ Hz, $f_{\max} = 8000$ Hz. Log-mel features are widely used in speech tasks because the mel scale compresses frequency in a perceptually meaningful way and the log transform stabilizes dynamic range [11]. This representation preserves *time–frequency structure* (e.g., energy contours, voicing-related harmonics, and spectral tilt), which is important for emotion cues such as prosody and intensity patterns. Spectrograms are padded/cropped to a fixed number of frames to enable mini-batch training and consistent input shapes for convolutional or sequence models.

(2) Mel/MFCC summary statistics. For classical baselines, we compute mel/MFCC-style features and aggregate over time using mean and standard deviation, producing a fixed-length vector per utterance. MFCCs compactly represent the short-term spectral envelope and are a strong baseline for speech classification when paired with linear or tree-based models [12]. The summary-statistics pooling discards fine-grained temporal order, but yields a simple and robust representation that is less sensitive to small timing variations and allows efficient training with limited data [12]. This provides a meaningful comparison point against neural models that directly consume time-resolved features.

D. Experimental Setup

Table I summarizes key implementation settings used in our runs.

Metrics. Overall accuracy and weighted F1 are reported on the validation and test splits. Although accuracy summarizes the proportion of correctly classified utterances, it may obscure class-specific failures when performance varies across emotions or when splits are imbalanced. Weighted F1 complements accuracy by averaging per-class F1 scores, which are the harmonic mean of precision and recall, weighted by class support. This approach penalizes models that perform well on frequent classes but poorly on less common ones. This is relevant for speech emotion recognition, where certain emotions, such as neutral, are often easier to classify and may dominate correct predictions. In contrast, confusable classes, such as sad versus neutral, may experience reduced recall. For the CNN-BiLSTM model, weighted F1 was not recorded in early runs and is therefore reported as missing in Table II.

TABLE II
PERFORMANCE ON CREMA-D USING SPEAKER-DISJOINT SPLITS.

Model	Val Acc	Test Acc	Val F1	Test F1
Mel+MFCC LR	0.49	0.46	0.49	0.46
CNN–BiLSTM	0.63	0.54	—	—
Wav2Vec2+LR	0.65	0.59	0.65	0.59
Ensemble ($\alpha = 0.25$)	**0.70**	**0.62**	**0.70**	**0.62**

IV. MODELS

A. Baseline: Logistic Regression

We train a multinomial Logistic Regression classifier on fixed-length feature vectors derived from Mel/MFCC summary statistics. Features are standardized and the model is trained with L2 regularization.

B. CNN–BiLSTM on Log-Mel Spectrograms

Our deep spectrogram model treats a log-mel spectrogram as a 2D input to convolutional blocks. Convolutions capture local spectro-temporal patterns (formants, harmonics, energy bursts), and a BiLSTM aggregates over time to model longer emotion dynamics. A final linear layer outputs class logits.

C. Compact Spectrogram Transformer

Inspired by transformer audio classifiers [8], we implement a small spectrogram transformer: spectrogram patches are embedded into tokens with positional encodings, processed by stacked self-attention layers, and pooled for classification.

D. wav2vec 2.0 Embeddings + Lightweight Classifier

We extract frame-level wav2vec 2.0 embeddings from a pretrained model [2] and pool them (mean and standard deviation pooling) into a single utterance embedding. A Logistic Regression classifier is then trained on these embeddings.

E. Ensemble

We form a linear ensemble by averaging predicted class probabilities from the CNN–BiLSTM and wav2vec 2.0 pipelines:

$$p_{\text{ens}} = \alpha\, p_{\text{cnn}} + (1-\alpha)\, p_{\text{w2v}},$$

with $\alpha = 0.25$ in our best run (25% CNN–BiLSTM and 75% wav2vec 2.0).

V. RESULTS

Table II summarizes results on our speaker-disjoint split. wav2vec 2.0 embeddings provide the strongest single-model generalization to the test speakers. The ensemble further improves test performance, suggesting the CNN–BiLSTM captures complementary spectrogram-local cues not fully represented in the pooled wav2vec 2.0 embedding.

Table III clarifies what each experiment changes: input representation, trainable capacity, and the main advantage of each approach. This also explains why the ensemble helps: wav2vec 2.0 contributes speaker-robust embeddings, while the CNN–BiLSTM contributes time–frequency patterns that are not fully captured by simple embedding pooling.

TABLE III
SUMMARY OF IMPLEMENTED SER PIPELINES AND WHAT EACH MODEL LEARNS.

Pipeline	Input	Trainable components	Key strength
Mel+MFCC LR	Summary stats	Logistic Regression head	Fast, interpretable baseline
CNN–BiLSTM	Log-mel (fixed frames)	CNN blocks + BiLSTM + classifier	Captures local + temporal cues
wav2vec 2.0 + LR	SSL embeddings (pooled)	Logistic Regression head	Strong transfer to unseen speakers
Ensemble ($\alpha = 0.25$)	Probabilities	Weighted averaging	Combines complementary cues

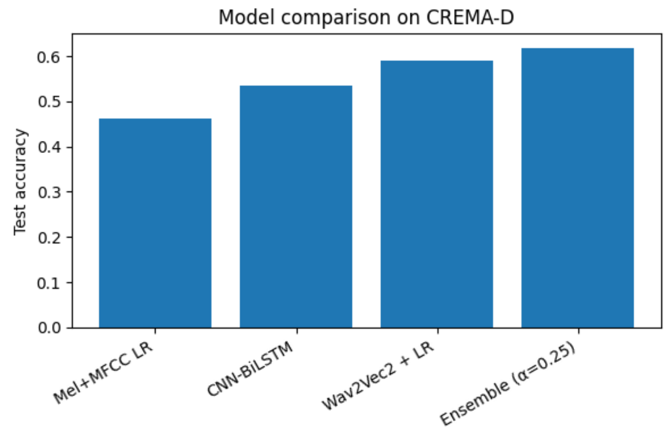

Fig. 2. Test accuracy comparison across models on CREMA-D (speaker-disjoint).

Figure 2 visualizes the same comparison on the held-out test speakers. The plot makes the trend easy to see: the handcrafted Mel/MFCC baseline is the weakest, the CNN–BiLSTM improves accuracy by modeling spectrogram dynamics, wav2vec 2.0 embeddings transfer better to unseen speakers, and the ensemble achieves the best test performance by combining complementary information from both pipelines.

A. Error Analysis

Figure 3 shows the ensemble confusion matrix. The largest off-diagonal errors are concentrated in a few class pairs, indicating systematic confusions rather than random noise. Using the label order (ANG, DIS, FEA, HAP, NEU, SAD), the ensemble most frequently confuses DIS$\rightarrow$SAD (36 samples) and SAD$\rightarrow$FEA (39 samples), and it also confuses HAP$\rightarrow$FEA (34 samples). These patterns motivate targeted improvements such as prosody-aware features (pitch/energy trajectories), stronger augmentation, and fine-tuning of SSL representations (Section VI).

B. Per-Class Performance

While overall accuracy is useful, SER systems often fail on specific emotions that are acoustically similar. Table IV

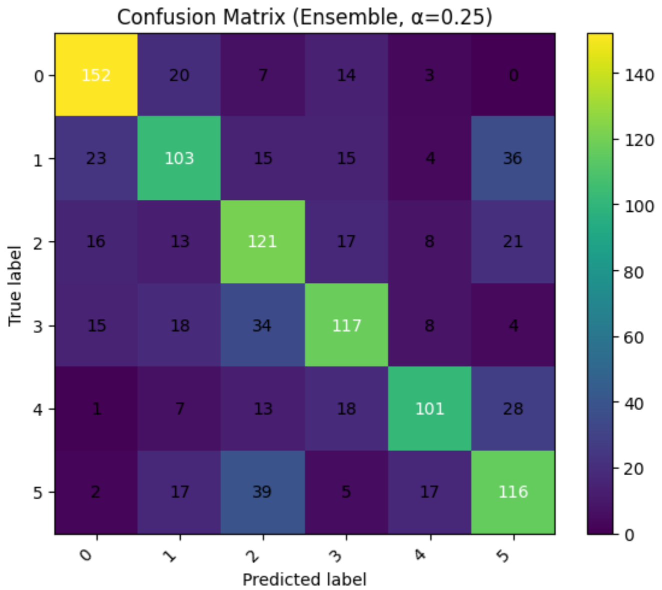

Fig. 3. Confusion matrix for the Ensemble model ($\alpha = 0.25$) on the speaker-disjoint split. Label order: ANG, DIS, FEA, HAP, NEU, SAD.

TABLE IV
PER-CLASS PERFORMANCE OF THE ENSEMBLE ($\alpha = 0.25$) ON THE SPEAKER-DISJOINT TEST SPLIT.

Emotion	Support	Prec.	Rec.	F1
ANG	196	0.727	0.776	0.751
DIS	196	0.579	0.526	0.551
FEA	196	0.528	0.617	0.569
HAP	196	0.629	0.597	0.613
NEU	168	0.716	0.601	0.654
SAD	196	0.566	0.592	0.579

reports per-class precision/recall/F1 for the ensemble on the speaker-disjoint test set, computed from the confusion matrix in Fig. 3. We observe the strongest performance for ANG and NEU, while DIS, FEA, and SAD remain the most confusable classes—consistent with the dominant off-diagonal errors discussed earlier.

Summary. The ensemble reaches 0.618 test accuracy, with macro-F1 ≈ 0.619 and weighted F1 ≈ 0.618.

VI. DISCUSSION AND FUTURE WORK

Our experiments demonstrate that enforcing a speaker-disjoint protocol substantially increases difficulty but yields a more realistic estimate of deployment performance on unseen speakers. Under this setting, the probability-averaging ensemble achieves 61.9% test accuracy with 0.619 weighted F1, outperforming both the handcrafted-feature baseline and each single-model pipeline. The results suggest that self-supervised wav2vec 2.0 embeddings provide strong speaker-robust representations, while the CNN–BiLSTM contributes complementary time–frequency cues that improve performance when combined.

A notable observation is the generalization gap for the CNN–BiLSTM (0.633 validation vs. 0.535 test accuracy), indicating sensitivity to speaker- or sentence-specific artifacts despite early stopping. In contrast, wav2vec 2.0 generalizes more consistently across held-out speakers. The confusion matrix in Fig. 3 indicates systematic confusions—most prominently DIS→SAD and SAD→FEA—which likely reflect overlap in prosodic and spectral patterns among these emotions in acted speech.

A. Limitations

This project evaluates audio-only emotion recognition on an acted dataset, so performance may not transfer directly to spontaneous real-world speech (different noise, speaking styles, and emotional expressions). Additionally, speaker-disjoint splitting improves scientific validity but reduces the amount of data per speaker seen during training, increasing variance and making hyperparameters more sensitive. Finally, our wav2vec 2.0 pipeline uses pooled frozen embeddings, which is intentionally lightweight but may under-utilize the representational capacity of SSL models without fine-tuning. [4].

B. Implications

These findings support two practical takeaways. First, speaker-disjoint evaluation should be treated as the default for SER reporting in order to avoid optimistic estimates caused by leakage. Second, combining heterogeneous representations (spectrogram-based and SSL-embedding-based) is an effective strategy for improving robustness without increasing training complexity dramatically.

C. Reproducibility Checklist

To ensure the results are auditable and repeatable, we: (i) fix random seeds and store the exact speaker-ID split lists, (ii) log all preprocessing parameters (sample rate, mel bins, max frames, padding/cropping policy), (iii) store trained model checkpoints and inference scripts for each pipeline, and (iv) report both aggregate metrics and class-level error patterns via confusion matrices.

D. Future Work

While the current system is complete and reproducible, several extensions are expected to yield additional gains, particularly under speaker-disjoint evaluation. (1) **Data augmentation:** apply SpecAugment-style time/frequency masking [13], additive noise, and random gain to reduce reliance on channel and speaker idiosyncrasies. (2) **Temporal pooling improvements:** replace simple mean/std pooling with attention-based pooling to better emphasize emotionally salient frames, consistent with prior SER work [7]. (3) **Stronger SSL adaptation:** fine-tune upper wav2vec 2.0 layers on CREMA-D with conservative learning rates and early stopping to align representations with emotion-relevant cues [2], and compare against HuBERT-style embeddings [9]. Finally, reporting additional metrics such as per-class F1 and calibration error would provide a more complete characterization of model behavior beyond overall accuracy.

VII. Conclusion

This project built a complete audio-only SER pipeline on CREMA-D with a strong emphasis on speaker-independent evaluation. We implemented and compared classical feature baselines, a CNN–BiLSTM spectrogram model, a compact spectrogram transformer, and transfer learning using wav2vec 2.0 embeddings. Our best current approach is a simple ensemble of wav2vec 2.0 and CNN–BiLSTM probabilities, reaching 69.9% validation accuracy and 61.9% test accuracy on a speaker-disjoint split. Future work will focus on augmentation, attention-based pooling, SSL fine-tuning, and targeted error reduction informed by confusion-matrix analysis.

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